

COM 3110/4115:

Text Processing

Text Encoding

Rob Gaizauskas

Department of Computer Science
University of Sheffield

- Languages and Writing Systems
- Characters vs Glyphs
- The Challenge of Character Encoding
- A Note on Binary/Octal/Hexadecimal Representations
- A (Very) Brief History of Character Encoding

Languages and Writing Systems

- Physical faculties required for **spoken language**
 - ◇ Broca's area, region of brain associated with language
 - ◇ vocal apparatus

arose in homo sapiens between 2 million and 300,000 years ago
- **Written language** only emerged between 3,500-3,100 BCE in 3 independent centres:
 - ◇ amongst the Harappans in the Indus Valley (Indus language)
 - ◇ amongst the Sumerians in Mesopotamia (cuneiform)
 - ◇ amongst the Egyptians (hieroglyphics)

though some dispute about whether earlier evidence counts as writing

Languages and Writing Systems (cont)

- Not an accident that the emergence of written language coincides with the stunning acceleration of human culture, science, technology and population size over the last 3000-4000 years.
 - ◇ “Humankind is defined by language; but civilization is defined by writing.” (Daniels and Bright, 1996)
- Today 7,168 languages **spoken** depending on, e.g. distinction made between language and dialect
 - ◇ just 23 languages are spoken by over half the world's population
 - ◇ over 40% of the world's languages are endangered (many with < 1,000 speakers)

(see <https://www.ethnologue.com/guides/how-many-languages>)
- However, only something like 370 **writing systems** of which only about 35 are in active use (according to omniglot.com)
 - ◇ Not clear how many languages lack writing systems, though only around 4,000 are known to have a written form
(see <https://omniglot.com/writing/stats.htm>)

Languages and Writing Systems (cont)

- What is a **writing system** (also called a **script**)?
“a system of more or less permanent marks used to represent an utterance in such a way that it can be recovered more or less exactly without the intervention of the utterer”

Peter T. Daniels, *The World's Writing Systems*

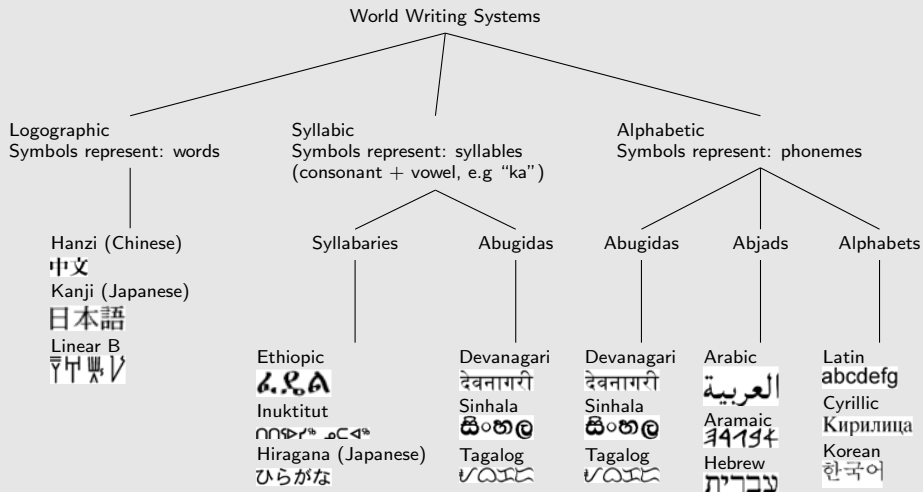
Languages and Writing Systems (cont)

- There is a **many-to-many mapping** between languages and scripts
 - ◇ many languages may share one script
 - e.g. most of the languages of Western Europe use the Roman script with some small variations (accents in French, some extra characters in Spanish, the Scandinavian languages)
 - ◇ some languages may have multiple scripts
 - Japanese – kanji (from Chinese); hiragana (for grammatical particles); katakana (for loan words from languages other than Chinese); arabic numerals

Languages and Writing Systems (cont)

- Writing systems may be classified along a number of dimensions:
 - ◇ **Directionality**
 - left-to-right – e.g. English, Russian
 - right-to-left – e.g. Arabic, Hebrew
 - top-to-bottom – e.g. Chinese
 - bottom-to-top – e.g. Mongolian
 - boustrophedon (“ox-turning”) – e.g. (some) ancient Greek
 - ◇ **Historical derivation**
 - ◇ **Relationship between symbols and sounds**
 - *phonological* systems show a clear relationship between sounds of the language and symbols
 - *non-phonological* systems do not
- **Sound/symbol relationship** generally agreed to be the best way to classify languages
 - ◇ but no consensus among scholars as to the best set of sub-classifications

Taxonomy of Writing Systems



- Languages and Writing Systems
- Characters vs Glyphs
- The Challenge of Character Encoding
- A Note on Binary/Octal/Hexadecimal Representations
- A (Very) Brief History of Character Encoding

Characters vs Glyphs

- A **character** is the smallest component of a writing system that has a semantic value.
 - ◇ I.e. changing one character to another – e.g. bat → cat – changes the meaning of the word in which it occurs
 - ◇ The word **grapheme** (by analogy with **phoneme**, the smallest sound unit in spoken language) is used
 - sometimes synonymously with “character”
 - sometimes to indicate multiple characters that function like a character – e.g. á
- A **glyph** is a representation of a character or characters, as it/they is/are rendered or displayed.
 - ◇ E.g. **A**, *A*, *Œ* are different glyphs representing the Latin character A
 - ◇ A repertoire of glyphs of similar appearance makes up a **font**
 - ◇ Changing font does **not** change the basic meaning of the word(s) whose font is changed
 - Though changing font to italics or bold can change, e.g., emphasis, which is an aspect of meaning

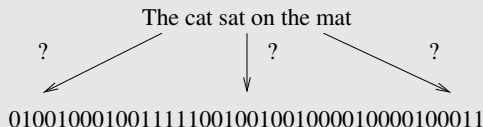
Characters vs Glyphs (Cont)

- The relationship between **glyphs** and **characters** can be complex
 - ◇ for any character there may be many glyphs representing it (in different fonts)
 - ◇ a single glyph may correspond to a number of characters.
 - E.g. “fi” is typically a single glyph used to represent “f” followed by “i” in typesetting English
 - ◇ there may be arbitrariness – should é
 - be stored as two characters and rendered using two glyphs?
 - be stored as two characters and rendered by one glyph?
 - be stored as one character and rendered by one glyph?
- When deciding on the set of characters for a language for purposes of representing them in a computer, it is extremely important (e.g. for sorting, indexing purposes) to separate out characters from glyphs
 - ◇ the underlying representation of a text should contain the character sequence only
 - ◇ the final appearance of the text is the responsibility of the rendering process

- Languages and Writing Systems
- Characters vs Glyphs
- The Challenge of Character Encoding
- A Note on Binary/Octal/Hexadecimal Representations
- A (Very) Brief History of Character Encoding

The Challenge of Character Encoding

- All data held in a digital computer is ultimately stored in binary form – i.e. as sequences of 0's and 1's
- Therefore, to store character data, i.e. text, in a digital computer it is necessary to agree an **encoding scheme**



- But what character encoding should we choose? What considerations should we bear in mind?

The Challenge of Character Encoding

- Various issues arise in designing a character encoding scheme:
 - ◇ **Generality** How many languages are you aiming to deal with?
 - ◇ **Character Set Specification** What is the character set(s) to be encoded?
 - ◇ **Hardware Issues** What are the minimal units of addressable memory?
 - ◇ **Variable/Fixed Width** Should every character occupy the same number of bits?
Or should more frequently occurring characters occupy fewer?
 - ◇ **Interoperability** Who do you want to play with? Do they already have a different scheme? (**standards**)

- Languages and Writing Systems
- Characters vs Glyphs
- The Challenge of Character Encoding
- A Note on Binary/Octal/Hexadecimal Representations
- A (Very) Brief History of Character Encoding

A Note on Binary, Octal & Hexadecimal Representations

- One byte = 8 bits.
- 8 bits can represent $2^8 = 256$ distinct values – 0 - 255.
- In binary notation, these range from

00000000
00000001
00000010
⋮
11111111

- In octal (base 8) notation, they range from: 000 to 377.
 - ◇ $o377 = (3 * 64) + (7 * 8) + 7$.
- In hexadecimal (base 16) notation, they range from 00 to *FF*
 - ◇ $xFF = (F * 16) + F$.
- Two bytes = 16 bits, and can represent $2^{16} = 65,536$ distinct values.
- Two byte values are most commonly written in hexadecimal; range from 0000 - *FFFF*
 - ◇ $d65535 = xFFFF = (15 * 4096) + (15 * 256) + (15 * 16) + 15$

- Languages and Writing Systems
- Characters vs Glyphs
- The Challenge of Character Encoding
- A Note on Binary/Octal/Hexadecimal Representations
- A (Very) Brief History of Character Encoding

A (Very) Brief History of Character Encoding

- “Standards are a good thing; and the good thing about standards is that there are so many to chose from ...”
- **Telegraphy and the Morse and Baudot Codes**
 - ◇ Telegraph invented by Wheatstone + Cooke (UK), 1837
 - ◇ Samuel Morse (US) invents simpler system + code, 1838
 - Variable length code, based on 2 values – dot's or dashes

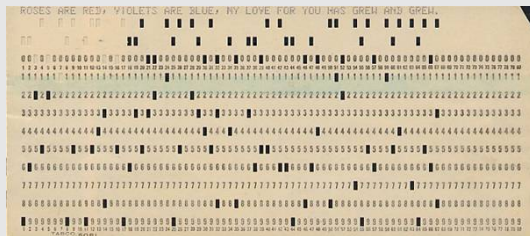
A	.-	N	-.	0	-----
B	-...-	O	---	1	.-....
C	-.-.-	P	.-.-	2	..-.-.
D	---.	Q	---.	3	-
E	.	R	.-.	4	-
F	...-	S	...-	5	
G	--.	T	-	6	-....
H		U	...-	7	-----
I	..	V	-	8	-----
J	-	W	...-	9	-----
K	---.	X	-.--	,	----- (comma)
L		Y	-.--	.	----- (period)
M	--	Z	----	?	-----

- ◇ Baudot (Fr) invents printing telegraph (“teleprinter”), 1874
 - used first binary code for textual data – 5 bit Baudot code
 - used a shift code to double code space; i.e., $2 * 2^5$

A (Very) Brief History of Character Encoding (cont)

• Hollerith and Punched Cards

- ◇ Herman Hollerith (US) invents punched card tabulating machines to process 1890 US census data
- ◇ code based on holes punched in 12 (row) × 80 (col) cards
- ◇ Hollerith commercialises invention – Tabulating Machine Co (1896) – becomes International Business Machines Corp (IBM) in 1924



From: http://wvegtter.hivemind.net/abacus/CyberHeroes/Hollerith_files/image006.jpg

A (Very) Brief History of Character Encoding (cont)

- **ASCII**

- ◇ American Standards Association (ASA – later changed to American National Standards Institute [ANSI]) proposed a 7-bit code – the **American Standard Code for Information Interchange** (ASCII) (1963, finalised 1968)

		ASCII																											
		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F												
0.	U0000	U0001	U0002	U0003	U0004	U0005	U0006	U0007	U0008	U0009	U000A	U000B	U000C	U000D	U000E	U000F													
	NUL	SOH	STX	ETX	EOT	ENQ	ACK	BEL	BS	HT	LF	VT	FF	CR	SO	SI													
1.	U0010	U0011	U0012	U0013	U0014	U0015	U0016	U0017	U0018	U0019	U001A	U001B	U001C	U001D	U001E	U001F													
	DLE	DC1	DC2	DC3	DC4	NAK	SYN	ETB	CAN	EM	SUB	ESC	FS	GS	RS	US													
2.	U0020	U0021	U0022	U0023	U0024	U0025	U0026	U0027	U0028	U0029	U002A	U002B	U002C	U002D	U002E	U002F													
	SP	!	"	#	\$	%	&	'	(	)	*	+	,	-	.	/													
3.	U0030	U0031	U0032	U0033	U0034	U0035	U0036	U0037	U0038	U0039	U003A	U003B	U003C	U003D	U003E	U003F													
	0	1	2	3	4	5	6	7	8	9	:	:	<	=	>	?													
4.	U0040	U0041	U0042	U0043	U0044	U0045	U0046	U0047	U0048	U0049	U004A	U004B	U004C	U004D	U004E	U004F													
	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O													
5.	U0050	U0051	U0052	U0053	U0054	U0055	U0056	U0057	U0058	U0059	U005A	U005B	U005C	U005D	U005E	U005F													
	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_													
6.	U0060	U0061	U0062	U0063	U0064	U0065	U0066	U0067	U0068	U0069	U006A	U006B	U006C	U006D	U006E	U006F													
	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o													
7.	U0070	U0071	U0072	U0073	U0074	U0075	U0076	U0077	U0078	U0079	U007A	U007B	U007C	U007D	U007E	U007F													
	p	q	r	s	t	u	v	w	x	y	z	{		}	~	DEL													

From: <http://www.gammon.com.au/unicode/>

- ◇ ASCII adopted by all US computer manufacturers except market-leaders IBM – proposed their own code – **Extended Binary Coded Decimal Interchange Code (EBCDIC)** – later abandoned
- ◇ Remains the most widely used coding scheme for English language text

A (Very) Brief History of Character Encoding (cont)

- **ISO/IEC 8859**

- ◇ ASCII insufficient for Western European languages
- ◇ International Standards Organisation proposed a derivative with 10 codes left for national variants – ISO 646 (1967)
- ◇ Has been rationalised and extended to an 8-bit code to cover most needs of Western and Eastern Europe – **ISO 8859** (revised up until 2001)
- ◇ Despite ISO 8859 many American PC companies built OS-specific extensions of ASCII for European languages that were not compatible with ISO 8859
- ◇ Not maintained or extended since 2001 – now subsumed by Unicode/UCS
- ◇ See https://en.wikipedia.org/wiki/ISO/IEC_8859.

- Paralleling develops in North America and Europe, efforts to develop standardised character codes have gone on in Asia

- ◇ for Chinese, Japanese and Korean (CJK) codes – e.g. JIS (Japanese Industrial Standard)
- ◇ for South Asian languages – e.g. ISCII (Indian Script Code for Information Interchange)

A (Very) Brief History of Character Encoding (cont)

- So far have discussed codes as developed to meet needs of individual countries. But may need to
 - ◊ process multiple languages in one document
 - ◊ reuse software for products dealing with multiple languages
- By mid-1980s it became clear there was a need for **multilingual coding schemes**
- Early efforts
 - ◊ some pioneering efforts by Xerox (Xerox Character Code Standard – XCCS) and IBM in early 1980's
 - ◊ TRON project (tronweb.super-nova.co.jp/homepage.html)
 - ongoing in Japan (U. Tokyo + Japanese industry) since 1984
 - uses escapes to shift between character planes – “limitlessly extensible”
 - built into BTRON operating system that runs on PCs

A (Very) Brief History of Character Encoding (cont)

- By late 1980's two major efforts underway to build character encoding schemes to embrace all the world's languages – past, present and future
 - ◊ **Unicode** started as initiative of US software industry in 1988
 - ◊ The **Universal Coded Character Set (UCS)** aka **ISO/IEC 10646** started in 1989 as an initiative of European and Japanese researchers

A (Very) Brief History of Character Encoding (cont)

- **Unicode** (www.unicode.org; en.wikipedia.org/wiki/Unicode)
 - ◇ committed to 16 bit codes and to avoiding escape sequences
 - ◇ supported by, e.g. Apple, IBM, Microsoft, Google, ...
 - ◇ now part of XML and HTML standards, Java, Perl, Python, ...
 - ◇ current version (Version 15.0.0, as of 25/09/22) contains a repertoire of 149,186 characters covering 161 modern and historic scripts, as well as multiple symbol sets
 - ◇ was initially criticised in East Asia for not meeting technical needs (character coverage insufficient; no way to expand without moving beyond 16 bit codes)
 - proponents: “the universal character encoding scheme for written characters and text” and “the foundation for global software”
 - opponents: “little more than an exercise in Western cultural imperialism”

A (Very) Brief History of Character Encoding (cont)

- **ISO/IEC10646** (en.wikipedia.org/wiki/Universal_Coded_Character_Set)
 - ◇ initially (1990) proposed a scheme involving 128 groups of 256 planes of 256 rows of 256 cells – could encode 679,477,248 characters
 - ◇ Supported four byte, two byte and variable length (1-5 byte) encodings
 - ◇ Was opposed by Unicode as too complex/requiring too much storage – wanted a two byte scheme
 - ◇ Later Unicode was forced to admit 16 bits was insufficient and added “surrogate pairs” (see next lecture)
- Unicode and UCS have now converged, are maintained together, and are code-for-code identical

Summary

- **Written language** only emerged in *homo sapiens* around 5,000-5,550 years ago and correlates strongly with the development of human civilization
- There are around 7000 languages spoken today, but only around 35 **writing systems** in wide use
- Writing systems differ along many dimensions, but are generally classified into three broad groupings according to the relation between sounds and symbols that they exhibit: **logographic**, **syllabic** and **alphabetic**
- A **character** is the smallest component of a writing system that has a semantic value; a **glyph** is a representation of a character or characters, as they are rendered or displayed
- Character encoding presents many challenges
- After more than 150 years of varying proposals for character encoding **Unicode** appears to be emerging as a global standard

References

- Crystal, D. The Cambridge Encyclopedia of Language. Cambridge University Press, 1987.
- Daniels, P. T. and Bright, W. eds, The World's Writing Systems. Oxford University Press, 1996.
- Steven J. Searle. "A Brief History of Character Codes in North America, Europe and Asia". Available at:
<http://tronweb.super-nova.co.jp/characcodehist.html>. Site last visited 25/09/22.
- Unicode Consortium. The Unicode Standard Version 15.0.0. Available at www.unicode.org. Site last visited 25/09/22.
- Unicode Technical Report # 17: Character Encoding Model. Available at www.unicode.org/reports/tr17. Site last visited 25/09/22.